Webb Telescope Finds an Ancient Galaxy Cluster That Grew Surprisingly Fast

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NASA's James Webb Space Telescope has taken a close look at a huge group of galaxies called XLSSC 122. What it found is puzzling scientists who study how the universe grew.

A **galaxy cluster** is a giant collection of galaxies held together by gravity. XLSSC 122 is about 10.4 billion **light-years** from Earth. Because its light took so long to reach us, we see it as it was roughly 10 billion years ago. Scientists call that time **cosmic noon**, when galaxies were making stars much faster than they do today.

Even at that early time, XLSSC 122 already looked large and organized, like clusters in today's universe. Webb's new images show that a lot of its mass is packed tightly into its center. Standard models say big structures should have grown more slowly, so this cluster does not match **predictions**.

The cluster also sits in front of galaxies that are even farther away. Its strong gravity bends their light into curved arcs, an effect called strong **gravitational lensing**. XLSSC 122 is now the most distant **galaxy cluster** known to show this effect. Scientists can use the bent light to measure how much mass the cluster has, including **dark matter**.

Dark matter is an invisible substance that does not give off light, but its gravity affects everything around it. Scientists estimate there is about five times more **dark matter** than ordinary matter. Most of the bending around XLSSC 122 comes from **dark matter**.

Kyle Finner, a scientist at IPAC, Caltech's astronomy data center, led the first of three studies about the cluster. He said strong lensing is a way to measure **dark matter** without actually seeing it.

Word Treasure

galaxy cluster -- A huge group of galaxies held together by gravity.

light-years -- A light-year is the distance light travels in one year, used to measure huge distances in space.

cosmic noon -- A time about 10 billion years ago when galaxies were making stars very quickly.

gravitational lensing -- When the gravity of a huge object bends light from something farther away, like a lens.

dark matter -- An invisible substance that gives off no light but pulls on things with its gravity.

predictions -- Statements about what scientists expect to happen or to find.

Background you need

The James Webb Space Telescope is a large space telescope that sees infrared light, which lets it study very distant and very old objects.

Because light takes time to travel, looking at something billions of light-years away shows how it looked billions of years ago.

Galaxy clusters are the largest groups of galaxies held together by gravity, and scientists use computer models to predict how fast they should grow.

Dark matter cannot be seen directly, so astronomers measure it through its gravity, such as the way it bends light from galaxies behind a cluster.

Break it down

LEAD: Webb studied an ancient galaxy cluster called XLSSC 122
KEY EVENT: The cluster's mass is packed tightly into its center
+- WHERE: About 10.4 billion light-years away
+- WHEN: Seen as it was about 10 billion years ago, during cosmic noon
PUZZLE: Standard models say big structures should have grown more slowly
EVIDENCE: Strong gravitational lensing bends light from farther galaxies into arcs
DARK MATTER: Most of the bending comes from invisible dark matter
OPEN QUESTION: Do scientists' models of how the universe grew need to change?

Why it matters

If clusters like this formed faster than expected, scientists may need to rethink how the universe built its largest structures.

Test what you remember

1. What is XLSSC 122?

- (A) A galaxy cluster
- (B) A space station
- (C) A comet
- (D) A new planet

2. About how far from Earth is XLSSC 122?

- (A) 4 light-years
- (B) 10 light-years
- (C) 1,000 miles
- (D) 10.4 billion light-years

3. Why does the cluster puzzle scientists?

- (A) A lot of its mass is packed into its center earlier than models predict
- (B) It has no galaxies
- (C) It is moving toward Earth
- (D) It is too cold

4. What is strong gravitational lensing?

- (A) A new type of camera lens
- (B) When strong gravity bends light from objects farther away
- (C) When stars explode
- (D) When a telescope zooms in

5. About how much more dark matter is there than ordinary matter?

- (A) About five times more
- (B) About half as much
- (C) The same amount
- (D) About 100 times more

6. Who led the first of three studies about the cluster?

- (A) Albert Einstein
- (B) A NASA astronaut
- (C) Edwin Hubble
- (D) Kyle Finner

Answer key (try the quiz first!): 1: A 2: D 3: A 4: B 5: A 6: D

Questions to think about

1. Kyle Finner, IPAC scientist: Says strong lensing is a way to measure dark matter without actually seeing it.

MY THOUGHTS (at least 20 words):